



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program:Bachelor of Technology
Semester : VIII

Branch: Computer Science and Engineering
Specialization: Computer Science and Engineering

Session: 2020-21

S.No.	Subject Code	Subject Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits
			Theory			Practical						
			End Sem.	Mid Sem. Exam.	Quiz/ Assignment	End Sem	Term work					
							Lab Work & Sessional					
1.	CS-81	Soft Computing	70	20	10	30	20	150	3	1	2	5
2.	CS-82	Cloud Computing	70	20	10	30	20	150	3	1	2	5
3.	CS-83	Elective-I	70	20	10	-	-	100	4	-	-	4
5.	CS-84	Major Project II	-	-		120	80	200	-	-	8	4
6.	CS-85	Project Seminar	-	-			100	100	-	-	6	3
7.	CS-86	Startup/Industrial Lectures(Internal assessment)					50	50			6	3
		Total	210	60	30	180	270	750	10	2	24	24

*End Semester Examination will be of three hours

ELECTIVE-I SUBJECT:-

1. CS -83 (A) Cyber Law & Ethics
3. CS-83 (C) Big Data

2. CS-83(B) Advance Computer Networks

PRACTICAL SUBJECT:-

- | | | |
|--------------------------|---------------------------------------|------------------------------|
| 1. CS- 81:SoftComputing | 2. CS-82 : CloudComputing | 3. CS-84 : Major Project -II |
| 4. CS-85:Project Seminar | 5. CS-86: Startup/Industrial Lectures | |

1 Hr Lecture	1 Hr Tutorial	2 Hr Practical
1 Credit	1 Credit	1 Credit



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF ENGINEERING**

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

**SYLLABUS OF COURSES TO BE OFFERED
“Bachelor of Technology”**

Disclaimer: The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026

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CS-81 Soft Computing

Unit I:- Introduction: Introduction to soft computing, application areas of soft computing, classification of soft computing techniques, structure & functioning of biological brain & Neuron, and concept of learning/training. Model of an Artificial Neuron, transfer/activation functions, perceptron, perceptron learning model, binary & continuous inputs, linear separability.

Unit II:- Multilayer Neural Networks: Feed Forward network - significance, training, loss function, Back-Propagation algorithm, convergence & generalization, momentum, applications. Feedback network -Hopfield Nets: architecture, energy functions, training algorithms & examples, competitive learning, self-organizing maps. Introduction to CNN and RNN network.

Unit III:- Fuzzy Systems: fuzzy set theory, fuzzy sets and operations, membership functions, concept of fuzzy relations and their composition, concept of fuzzy Measures. Fuzzy logic: fuzzy rules, inferencing. Fuzzy Control system: selection of membership functions, Fuzzification, rule based design & inferencing, defuzzification, applications of fuzzy system.

Unit IV:- Genetic algorithm: concepts, creation of offspring, working principle, encoding, fitness functions, reproduction, genetic modeling. Generation cycle & convergence of GA, application areas of GA.

Unit V:- Advanced soft computing techniques: Rough Set Theory - Introduction, Set approximation, Rough membership, Attributes, optimization. SVM - Introduction, obtaining the optimal hyper plane, linear and nonlinear SVM classifiers. Introduction to Swarm Intelligence, Swarm Intelligence Techniques: Ant Colony Optimization, Particle Swarm Optimization, Bee Colony Optimization etc.

LIST OF EXPERIMENTS:

1. Write A Program For Implementing Linear Saturating Function.
2. Study And Analysis Of Art Model.
3. Write A Program For Error Back Propagation Algorithm (Ebpa) Learning.
4. Study And Analysis Of CPN.
5. Study And Analysis Of Genetic Algorithm Life Cycle.
6. Study And Analysis Of Fuzzy Vs Crisp Logic.
7. Write A Program Of Perceptron Training Algorithm.
8. Write A Program To Implement Hebb's Rule.
9. Write A Program To Implement Of Delta Rule
10. Write A Program For Back Propagation Algorithm.
11. Write A Program To Implement Logic Gates



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REFERENCES

- 1) S.N. Sivanandam & S.N. Deepa, Principles of Soft Computing, Wiley Publications
- 2) S, Rajasekaran & G.A. Vijayalakshmi Pai, Neural Networks, Fuzzy Logic & Genetic Algorithms, Synthesis & applications, PHI Publication
- 3) Bose, Neural Network fundamental with Graph , Algo.& Appl, TMH Kosko: Neural Network & Fuzzy System, PHI Publication
- 4) Klir & Yuan ,Fuzzy sets & Fuzzy Logic: Theory & Appl., PHI Pub. Hagen, Neural Network Design, Cengage Learning





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CS-82 Cloud Computing

Unit-I Introduction: Historical development ,Vision of Cloud Computing, Characteristics of cloud computing as per NIST , Cloud computing reference model ,Cloud computing environments, Cloud services requirements, Cloud and dynamic infrastructure, Cloud Adoption and rudiments .Overview of cloud applications: ECG Analysis in the cloud, Protein structure prediction, Gene Expression Data Analysis ,Satellite Image Processing ,CRM and ERP ,Social networking.

Unit-II Cloud Computing Architecture: Cloud Reference Model, Types of Clouds, Cloud Interoperability & Standards, Scalability and Fault Tolerance, Cloud Solutions: Cloud Ecosystem, Cloud Business Process Management, Cloud Service Management. Cloud Offerings: Cloud Analytics, Testing Under Control, Virtual DesktopInfrastructure.

Unit –III Cloud Management & Virtualization Technology: Resiliency, Provisioning, Asset management, Concepts of Map reduce , Cloud Governance, High Availability and Disaster Recovery. Virtualization: Fundamental concepts of compute ,storage, networking, desktop and application virtualization .Virtualization benefits, server virtualization, Block and file level storage virtualization Hypervisor management software, Infrastructure Requirements , Virtual LAN(VLAN) and Virtual SAN(VSAN) and theirbenefits.

Unit-IV Cloud Security: Cloud Information security fundamentals, Cloud security services, Design principles, Secure Cloud Software Requirements, Policy Implementation, Cloud Computing Security Challenges, Virtualization security Management, Cloud Computing SecurityArchitecture.

Unit-V Market Based Management of Clouds , Federated Clouds/Inter Cloud: Characterization & Definition ,Cloud Federation Stack , Third Party Cloud Services . Case study : Google App Engine, Microsoft Azure , Hadoop , Amazon ,Aneka.

LIST OF EXPERIMENTS

1. Installation and configuration of Hadoop/Euceliptusetc.
2. Service deployment & Usage overcloud.
3. Management of cloudresources.
4. Using existing cloud characteristics & Service models.
5. Cloud Security Management.
6. Performance evaluation of services over cloud.

REFERENCES

- 1) Buyya, Selvi ,” Mastering Cloud Computing “,TMHPub
- 2) Kumar Saurabh, “Cloud Computing” , WileyPub
- 3) Krutz , Vines, “Cloud Security “ , WileyPub
- 4) Velte, “Cloud Computing- A Practical Approach” ,TMHPub
- 5) Sosinsky, “ Cloud Computing” , WileyPub



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CS-83 Elective-I (A) Cyber Law & Ethics

Unit I:- Introduction Computers and its Impact in Society, Overview of Computer and Web Technology, Need for Cyber Law, Cyber Jurisprudence at International and Indian Level, Cyber Law - International Perspectives UN & International Telecommunication Union (ITU) Initiatives Council of Europe - Budapest Convention on Cybercrime, Asia-Pacific Economic Cooperation (APEC), Organization for Economic Co-operation and Development (OECD), World Bank, Commonwealth of Nations.

Unit II:- Constitutional & Human Rights Issues in Cyberspace Freedom of Speech and Expression in Cyberspace, Right to Access Cyberspace – Access to Internet, Right to Privacy, Right to Data Protection, Cyber Crimes & Legal Framework Cyber Crimes against Individuals, Institution and State, Hacking, Digital Forgery, Cyber Stalking/Harassment, Cyber Pornography, Identity Theft & Fraud Cyber terrorism, CyberDefamation.

Unit III:- Cyber Torts Cyber Defamation, Different Types of Civil Wrongs under the IT Act 2000, Intellectual Property Issues in Cyber Space Interface with Copyright Law, Interface with Patent Law, Trademarks & Domain Names Related issues.

Unit IV:- E-Commerce Concept, E-commerce-Salient Features, Online approaches like B2B, B2C & C2C Online contracts, Click Wrap Contracts, Applicability of Indian Contract Act, 1872,

Unit V:- Dispute Resolution in Cyberspace, Concept of Jurisdiction, Indian Context of Jurisdiction and IT Act, 2000. International Law and Jurisdictional Issues in Cyberspace, Dispute Resolutions.

REFERENCES

- 1) Chris Reed & John Angel, Computer Law, OUP, New York.
- 2) Justice Yatindra Singh, Cyber Laws, Universal Law Publishing Co, New Delhi.
- 3) Verma S, K, Mittal Raman, Legal Dimensions of Cyber Space, Indian Law Institute.
- 4) Jonathan Rosenoer, Cyber Law, Springer, New York.
- 5) Sudhir Naib, The Information Technology Act, 2005: A Handbook, OUP, New York.
- 6) S.R. Bhansali, Information Technology Act, 2000, University Book House Pvt. Ltd.

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CS-83 Elective -I (B) Advance Computer Networks

UNIT I:- Review of Networking and O.S. fundamentals, ISO-OSI Model, different layers and their functions, LAN, MAN, WAN, Communication media & principles IEEE standards etc.

UNIT II:- Internetworking with TCP/IP, Basic concepts, Principles, Protocols and Architecture, Address handling Internet protocols and protocol layering. DNS, Applications: TELNET, RLOGN , FTP, TFTP, NFS, SMTP, POPL, IMAP, MIME, HTTP,STTP,DHCP, VOIP, SNMP.

UNIT III:- Introduction to Router, Configuring a Router, Interior & Exterior Routing, RIP, Distance Vector Routing, OSPF, BGP, Uni-cast, Multicast and Broadcast. Multicast routing protocols: DVMRP, MOSPF, CBT, PIM, MBONE, EIGRP, CIDR, Multicast Trees, Comparative study of IPv6 andIPv4.

UNIT IV:- VPN addressing and routing, VPN Host management, ATM Concepts, Services Architecture, Equipment's and Implementation

UNIT V:- Introduction to wireless transmission and medium access control, wireless LAN: IEEE 802.11, Hiper LAN , Bluetooth Mobile Network and Transport layer, WAP GSM and CDMA: Network architecture andmanagement

REFERENCES

1. Computer Networks:Tanenbaum.
2. Internetworking with TCP/IP:Comer.
3. Data Communications, Computer Networks and Open Systems:Hallsall.
4. Data Communications,Stalling.
5. Mobile Communication: Schiller, PearsonEducation.





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CS-83 Elective -I (C) Big Data

Unit I:- Introduction to Big data, Big data characteristics, Types of big data, Traditional versus Big data, Evolution of Big data, challenges with Big Data, Technologies available for Big Data, Infrastructure for Big data, Use of Data Analytics, Desired properties of Big Data system.

Unit II:- Introduction to Hadoop, Core Hadoop components, Hadoop Eco system, Hive Physical Architecture, Hadoop limitations, RDBMS Versus Hadoop, Hadoop Distributed File system, Processing Data with Hadoop, Managing Resources and Application with Hadoop YARN, MapReduce programming.

Unit III:- Introduction to Hive Hive Architecture, Hive Data types, Hive Query Language, Introduction to Pig, Anatomy of Pig, Pig on Hadoop, Use Case for Pig, ETL Processing, Data types in Pig running Pig, Execution model of Pig, Operators, functions, Data types of Pig.

Unit IV:- Introduction to NoSQL, NoSQL Business Drivers, NoSQL Data architectural patterns, Variations of NOSQL architectural patterns using NoSQL to Manage Big Data, Introduction to MangoDB

Unit V:- Mining social Network Graphs: Introduction Applications of social Network mining, Social Networks as a Graph, Types of social Networks, Clustering of social Graphs Direct Discovery of communities in a social graph, Introduction to recommender system.

REFERENCES

- 1) KaiHwang, GeoffreyC., Fox. Jack, J. Dongarra, "Distributed and Cloud Computing", Elsevier, First edition
- 2) Michael Minelli, Michele Chambers, AmbigaDhiraj, "Big Data BigAnalytics", Wiley
- 3) RadhaShankarmani, M. Vijaylakshmi, " Big Data Analytics", Wiley, Second edition
- 4) Seema Acharya, SubhashiniChellappan, " Big Data and Analytics", Wiley, First edition



CS-84 Major Project II

Objectives of the course Major Project are:

- ☐ To provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses.
- ☐ To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems.
- ☐ To give students an opportunity to do something creative and to assimilate real life work situation in institution.
- ☐ To adapt students for latest development and to handle independently new situations.
- ☐ To develop good expressions power and presentation abilities in students.

The focus of the Major Project is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write up i.e. detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration of the working system (if any)

Working schedule The faculty and student should work according to following schedule: Each student undertakes substantial and individual project in an approved area of the subject and supervised by a member of staff. The student must submit outline and action plan for the project execution (time schedule) and the same be approved by the concerned faculty. Action plan for Major Project work and its evaluation scheme



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Task/Process	Week	Evaluation	Marks For Term Work
Orientation of students by HOD/Project Guide	1st	-	-
Literature survey and resource collection	2nd	-	-
Selection and finalization of topic before a committee	3rd Seminar	I	10
Detailing and preparation of Project (Modeling, Analysis and Design of Project work	4th to 5th	-	10
Development stage		-	
Testing, improvements, quality control of	6th to 10th 11th	-	25
Acceptance testing	12th	-	10
Report Writing	13th to 15th	-	15
Presentation before a committee (including user manual, if any)	16th	Seminar-II	30

* Committee comprises of HOD, all project supervisions including external guide from industry (if any)

NOTE: At every stage of action plan, students must submit a write up to the concerned guide



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CS-85 Project Seminar

In Project Seminar students have to make a PPT of the selected topic of Project and present in front of the faculties & students. By doing this, in front of faculties & students, students communication skill will get developed & will be a good presenter. By doing this, students Project knowledge will also get enhanced.





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CS-86 Startup/ Industrial Lectures

Objective of Startup/Industrial Lectures:

The objective of undertaking industrial lectures is to provide work experience so that student's engineering knowledge is enhanced and employment prospects are improved. The student should take this as a window to the real World and should try to learn as much as possible from real life experiences by involving and interacting with industry staff. Industrial lectures also provides an opportunity to students to select an engineering problem and possibly an industry guide for their professional growth.

